Remember the following Conversion table:

Length	Weight	Volume (Capacity)
• 1 km = 1000 m • 1 m = 100 cm • 1 m = 1000 mm • 1 cm = 10 mm	• 1 kg = 1000 g • 1 g = 1000 mg	• 1 kl = 1000 l • 1 l = 1000 ml

Before we study Conversion of metric units, we should learn about decimal point position when multiplying / dividing of a decimal number by the powers of ten.

When a decimal number is multiplied by 10, 100, 1000, 10000, we move the decimal point by 1, 2 ,3 ,4 places to the right respectively.	**When a decimal number is divided by 10, 100, 1000, or 10000, we move the decimal point 1, 2, 3, 4 places to the left respectively.**
Example: **Multiply 345.972 by 10,100,1000 and 10000** 345.972 × 10 = 3459.72 (move the decimal point by one place to the right) 345.972 × 100 = 34597.2 (move the decimal point by two places to the right) 345.972 × 1000 = 345972 (move the decimal point by three places to the right) 345.9720 × 10000 = 3459720 (move the decimal point by four places to the right) Since, there are only 3 digits in the decimal part, add a zero to the right and then place the decimal point.	**Example :** **Divide 647.39 by 10,100 and 1000 and 10000** $\frac{647.39}{10} = 64.739$ (move the decimal point one place towards its left) $\frac{647.39}{100} = 6.4739$ (move the decimal point two places towards its left) $\frac{647.39}{1000} = 0.64739$ (move the decimal point three places towards its left) $\frac{0647.39}{10000} = 0.064739$ (move the decimal point four places towards its left) Since there are only 3 digits in the integral part , add one zero and place the point before it.

Example 1: The official distance of Marathon race is 42.195 km. Express this distance in metre.

Solution : The official distance of Marathon race is 42.195 km.

= 42.195 X 1000 m

= 42195 m

1 km = 1000 m

Example 2: The average rainfall of Tamilnadu is 998 mm. convert it into cm.

Solution : The average rainfall =998 mm = 998.0 x $\frac{1}{10}$ cm

$= \frac{998.0}{10}$ cm

= 99.8 cm

1 cm = 10 mm

$\frac{1}{10}$ cm = 1 mm

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